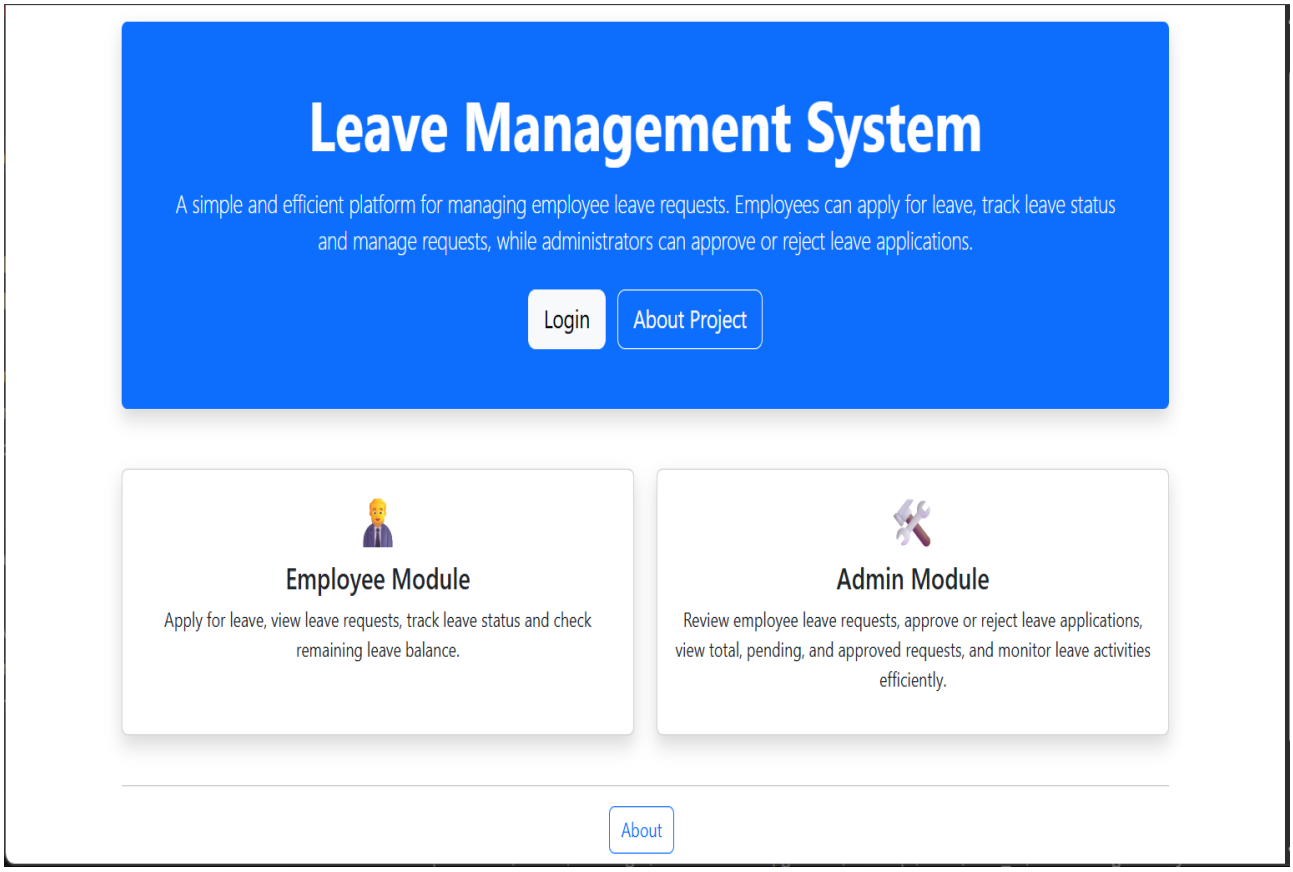
Project Name :-

Staff Leave Manager - Leave Management System



From:- Aditya Sunil Garud

Manager:- Poonam Pawar

ABSTRACT

Staff Leave Manager - Leave Management System is a mini project developed using ASP.NET Core MVC, Entity Framework Core, MySQL, and Bootstrap. The project is designed to provide a simple and efficient way of managing employee leave requests within an organization. The system helps reduce manual work and improves the overall leave management process by maintaining leave records in a centralized database.

The application allows employees to log in using their assigned credentials and perform various leave-related activities. Employees can apply for leave, view their submitted leave requests, track the status of applications, and check their available leave balance. This helps employees manage their leave information easily through a single web-based platform.

The system also includes an Admin Module that allows administrators to manage employee leave requests effectively. Administrators can view all leave applications, approve or reject requests, monitor leave activities, and maintain employee leave records through a dedicated dashboard. This helps ensure a smooth and transparent leave approval process within the organization.

The project is developed using ASP.NET Core MVC architecture, which provides better organization of application components through Models, Views, and Controllers. Entity Framework Core is used for handling database operations, while MySQL serves as the backend database for storing employee and leave-related information. Bootstrap is used to create a responsive and user-friendly interface.

The main objective of this mini project is to automate the leave management process and provide a centralized system for handling leave requests. Through the development of this project, practical knowledge of MVC Architecture, Authentication, Authorization, CRUD Operations, Database Management, and Software Development Life Cycle (SDLC) was gained. Overall, the Leave Management System provides a simple, secure, and effective solution for managing employee leave information in an organized manner.

Software Development Life Cycle (SDLC)

Waterfall SDLC Model

The Software Development Life Cycle (SDLC) is a structured process used for developing software applications. It divides the software development process into different phases and ensures that all project requirements are implemented successfully.

For the Leave Management System, the Waterfall Model was selected because the project requirements were clearly identified before the development phase started. The model follows a sequential approach where each phase is completed before moving to the next phase.

Waterfall Phases

- Planning
- Requirement Analysis
- System Design
- Development
- Testing
- Deployment
- Maintenance

Benefits of Using Waterfall Model for This Project

- Clear and well-defined development phases.
- Easy to understand and implement.
- Suitable for projects with fixed requirements.
- Provides proper documentation at each stage.
- Helps monitor project progress effectively.
- Reduces confusion during development.
- Ensures systematic implementation of project functionalities.

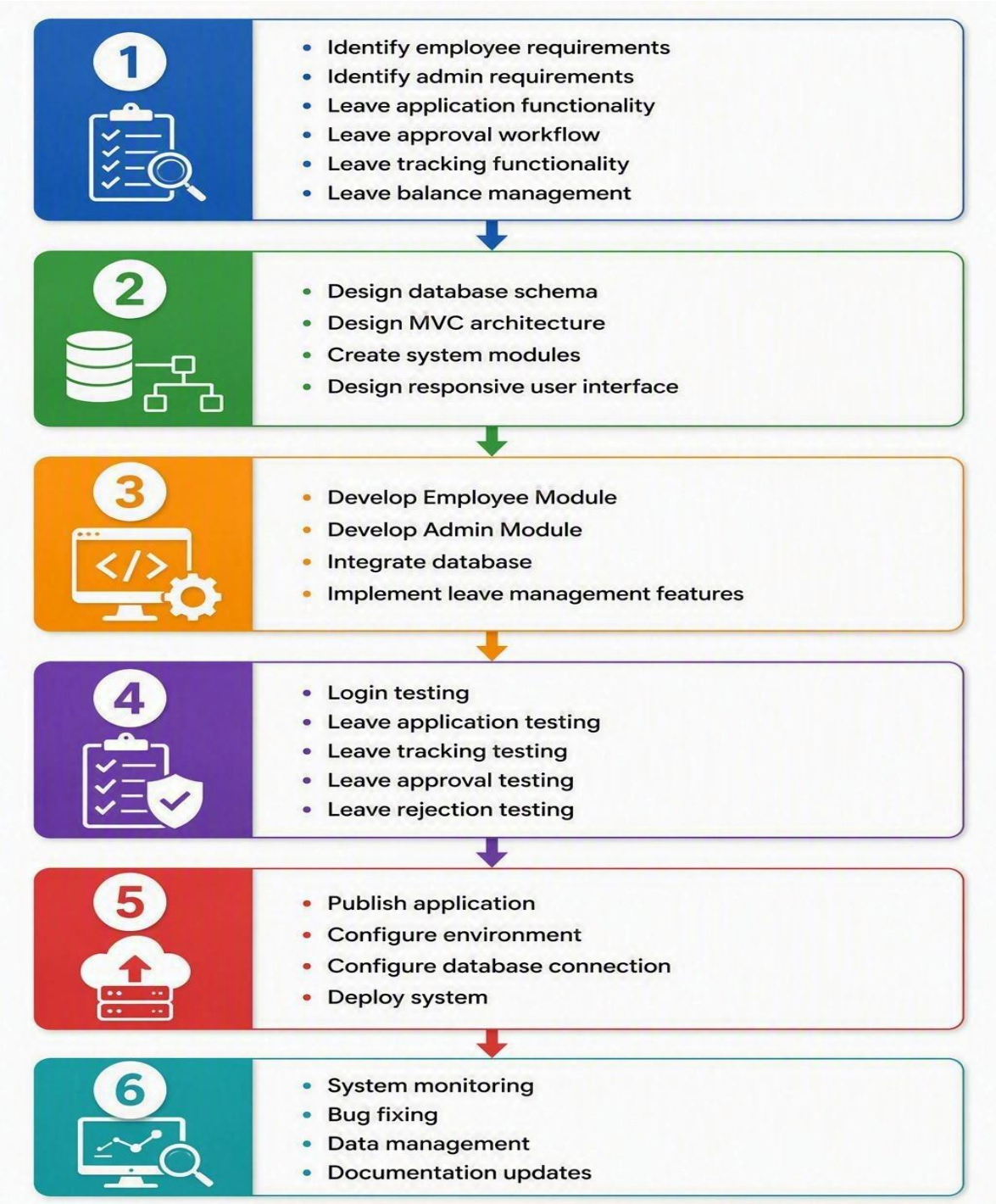


Figure 1: Waterfall SDLC Model Used for Staff Leave Manager - Leave Management System

The Waterfall Model provided a clear and systematic approach for developing the Leave Management System. Each phase was completed one after another, which helped in managing the project efficiently and ensuring that all requirements were properly implemented. It also made the development process easier to understand, monitor, and document

Planning and Requirement Analysis

The planning phase focused on identifying the requirements of employees and administrators involved in the leave management process. The main objective was to develop a centralized system that simplifies leave application, approval, and tracking activities.

Traditional leave management methods often involve manual records and paperwork, which can lead to errors and delays. The Leave Management System was developed to automate these activities and improve efficiency.

Employee Requirements

- Login using valid credentials
- Apply for leave
- View leave requests
- Track leave status
- View available leave balance
- Logout securely

Admin Requirements

- Login securely
- View all leave requests
- Approve leave applications
- Reject leave applications
- View leave request summary
- Monitor employee leave activities
- Logout securely

The requirements gathered during this phase served as the foundation for the development of the Leave Management System. Understanding the needs of both Employees and Administrators helped in designing a system that is simple, efficient, and capable of managing leave-related activities in an organized manner

Use Case Diagram

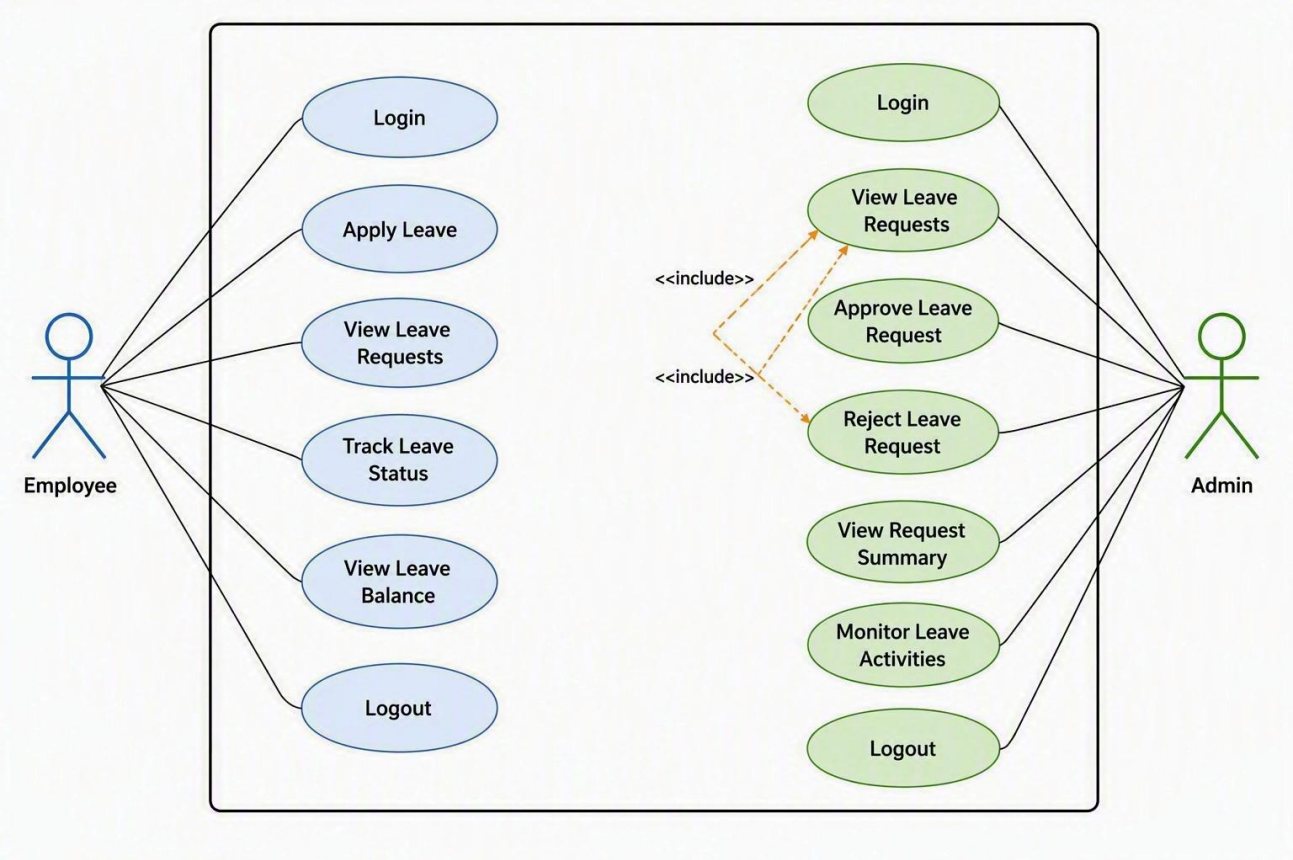


Figure 2: Use Case Diagram of for Staff Leave Manager - Leave Management System

The Use Case Diagram represents the interaction between the primary users of the Leave Management System and the functionalities available to them. The system consists of two main actors, namely Employee and Admin, each having specific roles and responsibilities within the leave management process. Employees can log in, apply for leave, view submitted leave requests, track leave status, and check their available leave balance. Administrators can log in, view all leave requests, approve or reject leave applications, monitor employee leave activities, and view leave request summaries. The diagram provides a high-level overview of the system requirements and clearly illustrates how different users interact with the application to perform leave management activities. It also helps in understanding the overall workflow and functionality of the Leave Management System.

System Design

The Leave Management System was designed using the ASP.NET Core MVC Architecture. This architecture separates the application into Models, Views, and Controllers, making the system easy to develop, manage, and maintain. The architecture provides proper separation of responsibilities and ensures smooth communication between the user interface, business logic, and database.

Employees and Administrators interact with the application through web pages developed using ASP.NET Core MVC and Bootstrap. User requests are processed by Controllers, which communicate with the business logic and database layer to perform the required operations. Entity Framework Core is used for database access, and MySQL is used as the backend database for storing employee and leave-related information.

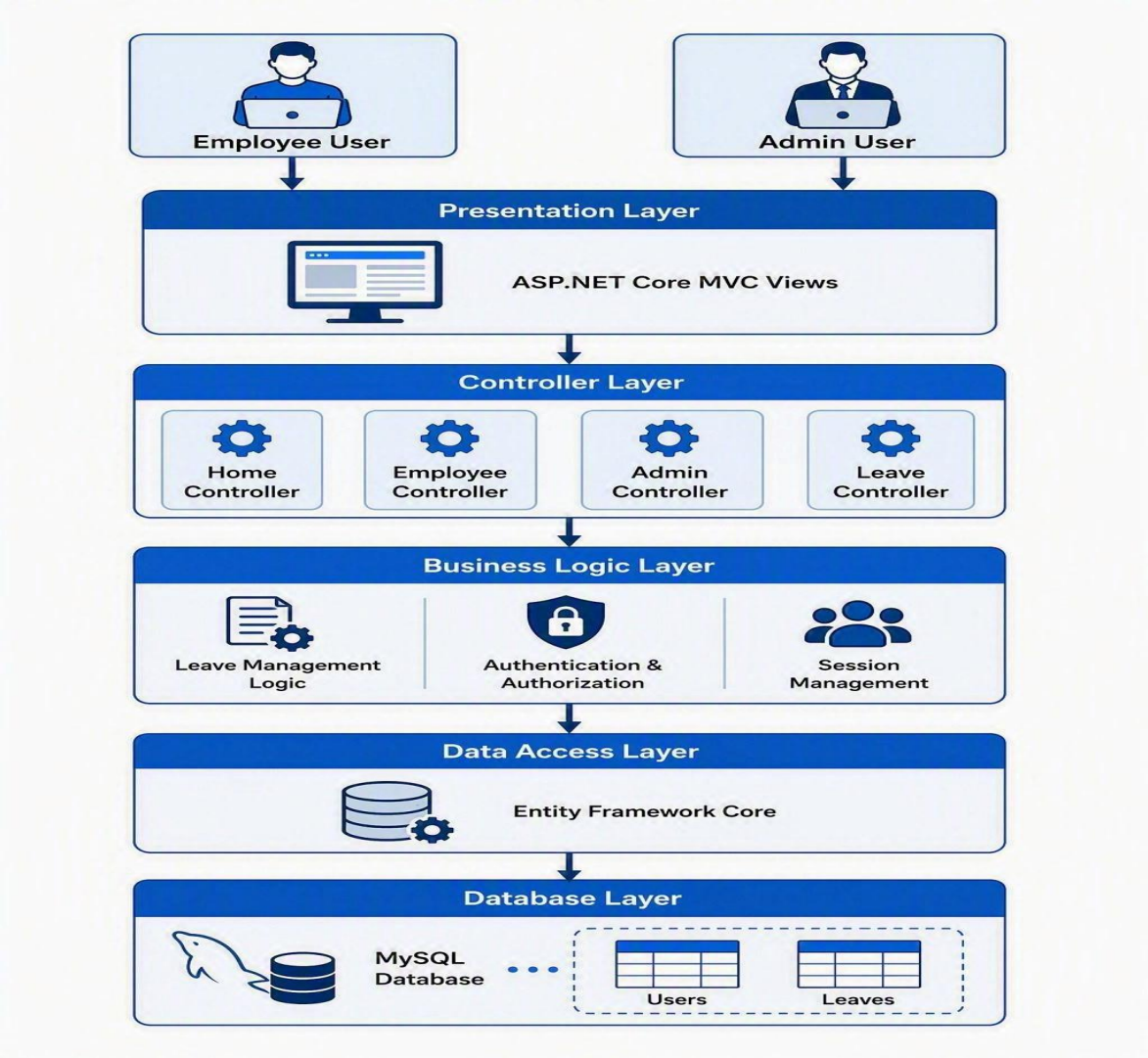


Figure 3: System Architecture Diagram of Staff Leave Manager - Leave Management System

Technologies Used

The Leave Management System was developed using modern web development technologies that provide security, flexibility, and efficient database management.

Technologies Used

- ASP.NET Core MVC
- C#
- Entity Framework Core
- MySQL
- Bootstrap
- HTML
- CSS
- Razor Views

ASP.NET Core MVC was used to develop the application following the MVC architecture pattern. Entity Framework Core was used to perform database operations and manage data efficiently. MySQL was used as the backend database, while Bootstrap was used to create a responsive and user-friendly interface.

The selected technologies provided a strong foundation for developing the Leave Management System. ASP.NET Core MVC helped in organizing the application using the Model-View-Controller architecture, making the system easier to manage and maintain. Entity Framework Core simplified database operations and reduced the amount of manual SQL code required for data access. MySQL ensured reliable storage and management of employee and leave-related information, while Bootstrap helped create a responsive and user-friendly interface. Together, these technologies enabled the development of a secure, efficient, and scalable web application that successfully meets the requirements of employee leave management.

Database Design

The database design of the Leave Management System was created to manage users and leave applications efficiently. A relational database structure was used to maintain data consistency and proper relationships between tables.

Users Table

The Users table stores information about Employees and Administrators.

Fields

- Id
- Name
- Email
- Password
- Role
- TotalLeaves
-

Leaves Table

The Leaves table stores leave request information submitted by employees.

Fields

- Id
- FromDate
- ToDate
- Reason
- Status
- UserId

Relationship

- One User can have multiple Leave Requests.
- Each Leave Request belongs to one User.
- UserId acts as a Foreign Key referencing the Users table.

The database design of the Leave Management System provides a structured and efficient way to store and manage employee and leave-related information. The Users table maintains details of both Employees and Administrators, including their login credentials, roles, and available leave balance. The Leaves table stores all leave applications submitted by employees, including leave dates, reasons, and approval status. A one-to-many relationship is established between the Users and Leaves tables, where one user can submit multiple leave requests, while each leave request belongs to a single user. This relational database structure ensures data consistency, simplifies record management, and supports efficient retrieval of information required for leave application and approval processes.

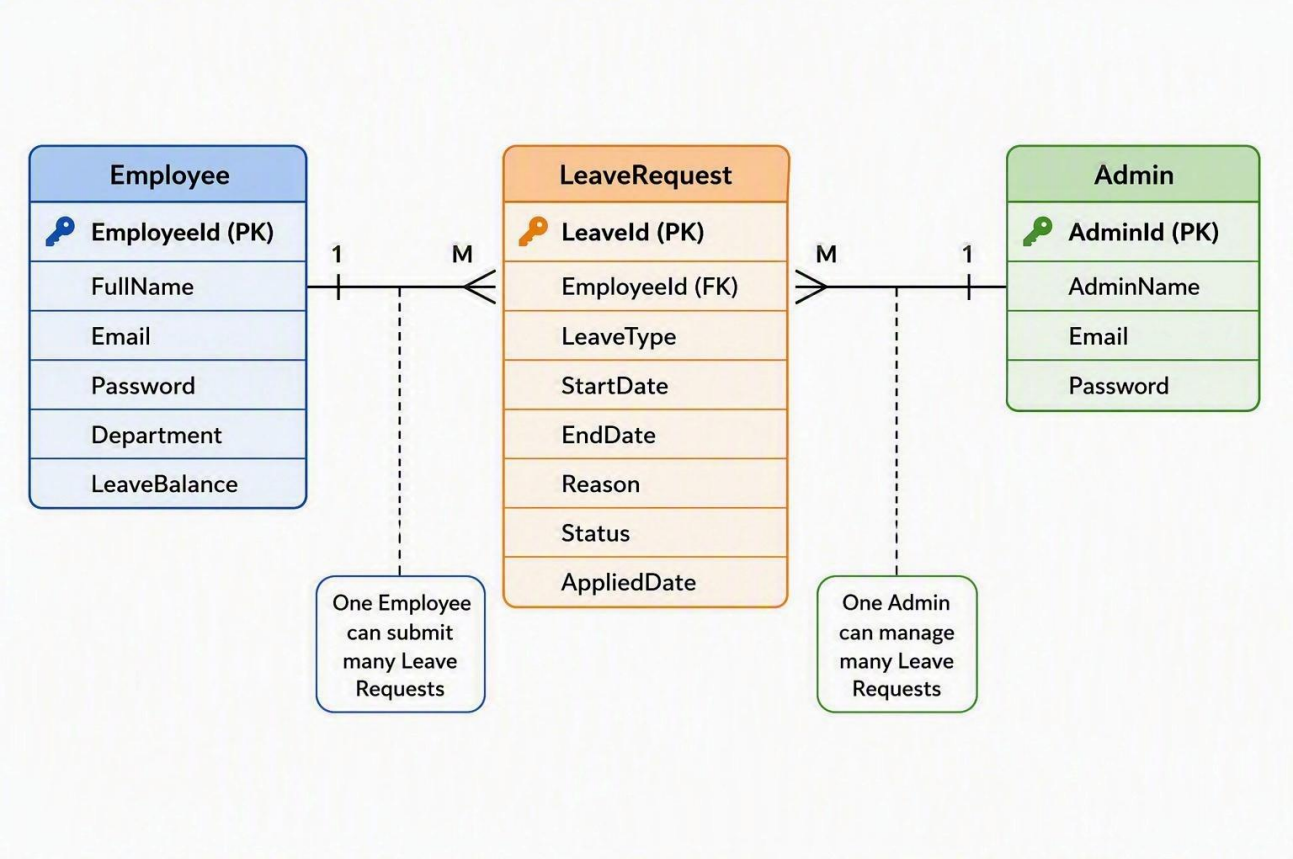


Figure 4: ER Diagram of Staff Leave Manager - Leave Management System

The ER Diagram represents the database structure of the Leave Management System and illustrates the relationship between the Users and Leaves tables. It provides a clear understanding of how employee information and leave request records are stored and managed within the system. The Users table stores details of Employees and Administrators, including their login credentials, roles, and available leave balance, while the Leaves table contains information related to leave applications such as leave dates, reasons, status, and user references.

A one-to-many relationship exists between the Users and Leaves tables, where one user can submit multiple leave requests and each leave request belongs to a single user. The UserId field in the Leaves table acts as a Foreign Key that references the Id field of the Users table, ensuring proper relationship management and data consistency. The ER Diagram serves as a visual representation of the database design and helps in understanding how different entities interact with each other to support leave application, approval, and tracking processes within the system.

Modules

Employee Module

The Employee Module provides all functionalities required by employees to manage leave applications. Employees can securely access the system and perform various leave-related activities through a user-friendly dashboard.

Employee Features

- Login
- Apply Leave
- View Leave Requests
- Track Leave Status
- View Leave Balance
- Logout

Screenshots

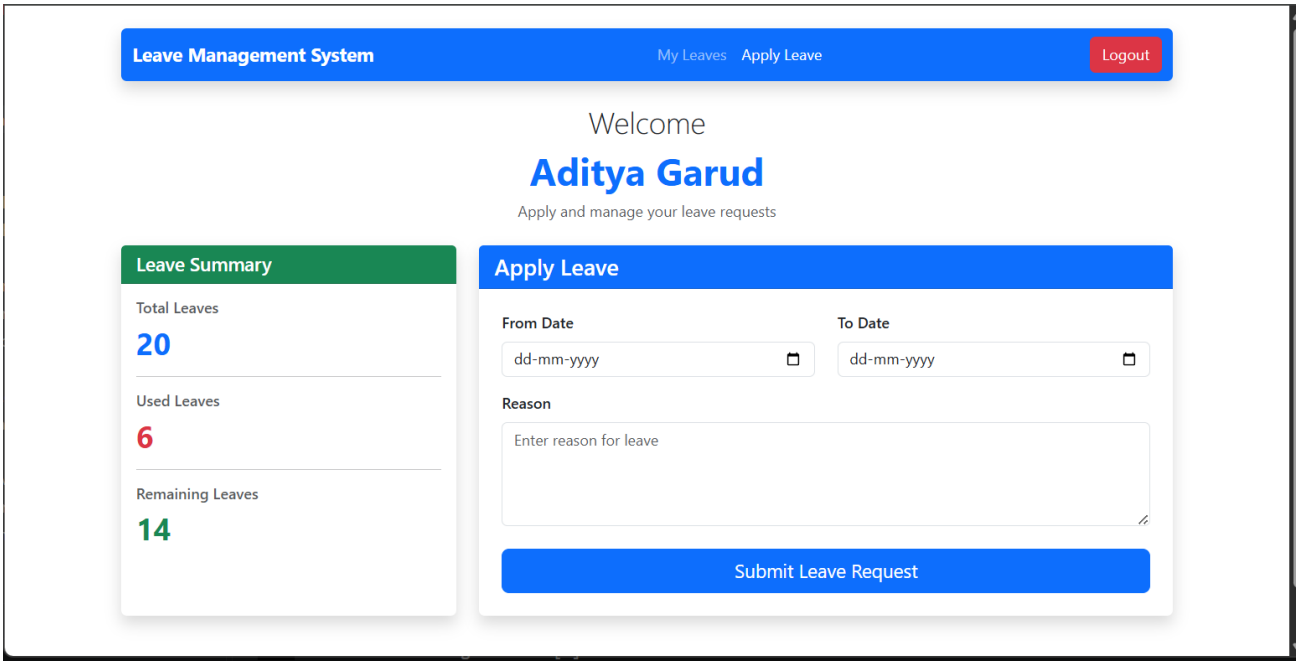


Figure 5: Employee Dashboard- Apply Leave Page

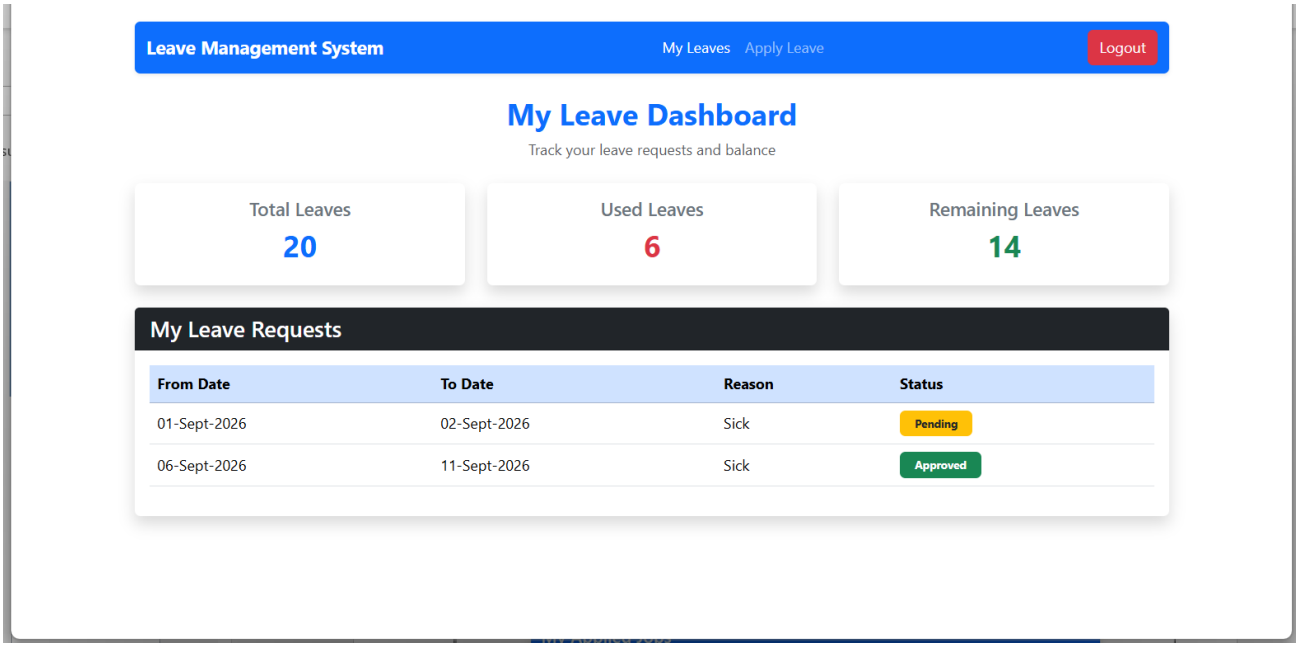


Figure 7: My Leaves dashboard

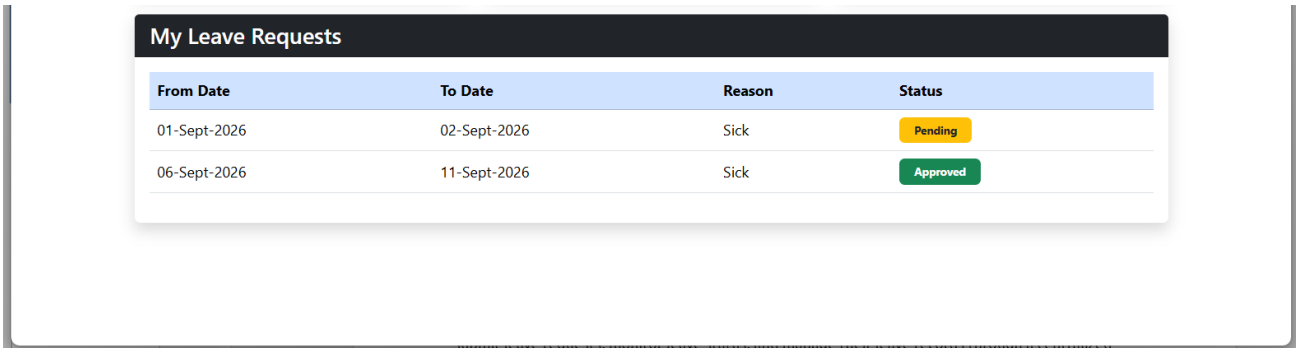


Figure 8: Leave history (Sub-Page)

The screenshots above demonstrate the functionalities available in the Employee Module. Employees can submit leave requests, monitor leave status, and manage their leave records through a centralized platform.

Admin Module

The Admin Module helps administrators manage employee leave activities efficiently. Administrators can access leave information, approve or reject requests, and monitor employee leave records through a dedicated dashboard.

Admin Features

- Login
- View Leave Requests
- Approve Leave Requests
- Reject Leave Requests
- View Request Summary
- Monitor Leave Activities
- Logout

Screenshots

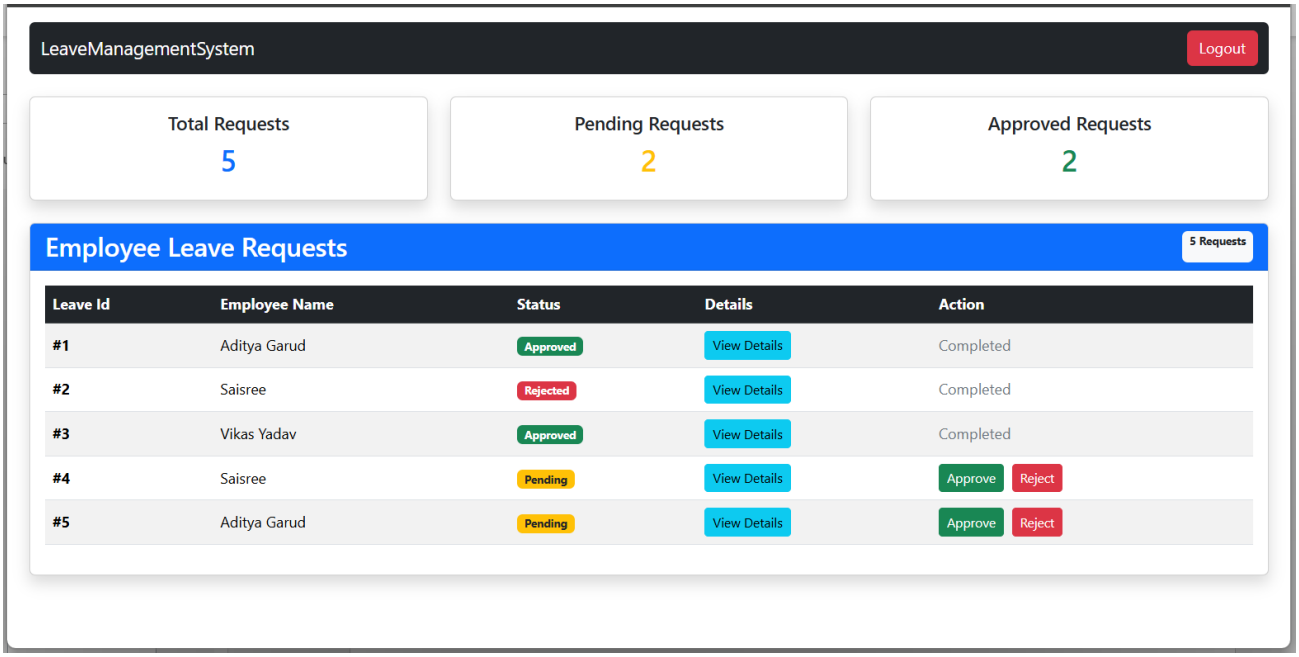


Figure 9: Admin Dashboard

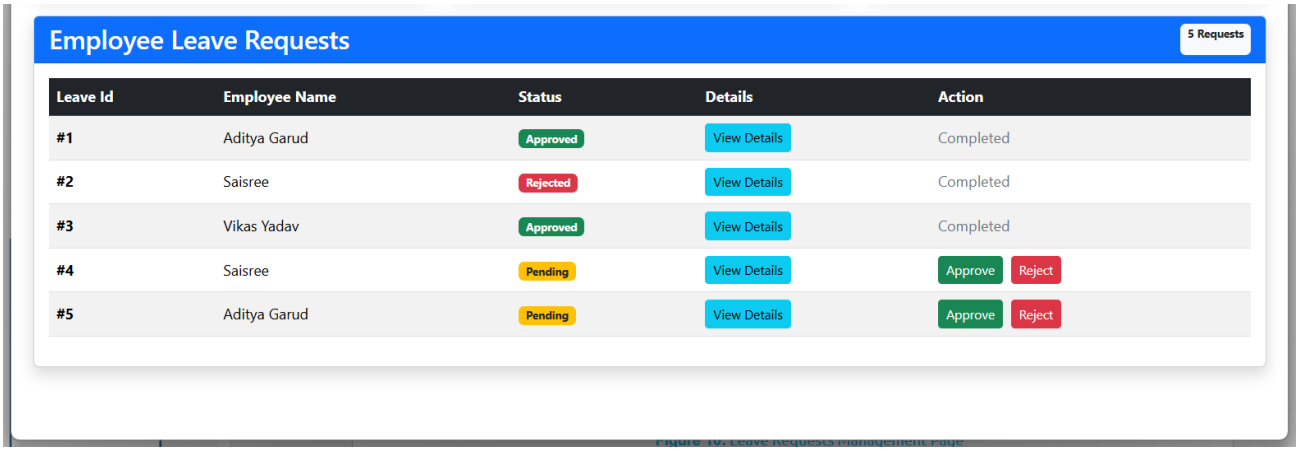


Figure 10: Leave Requests Management (Sub-Page)

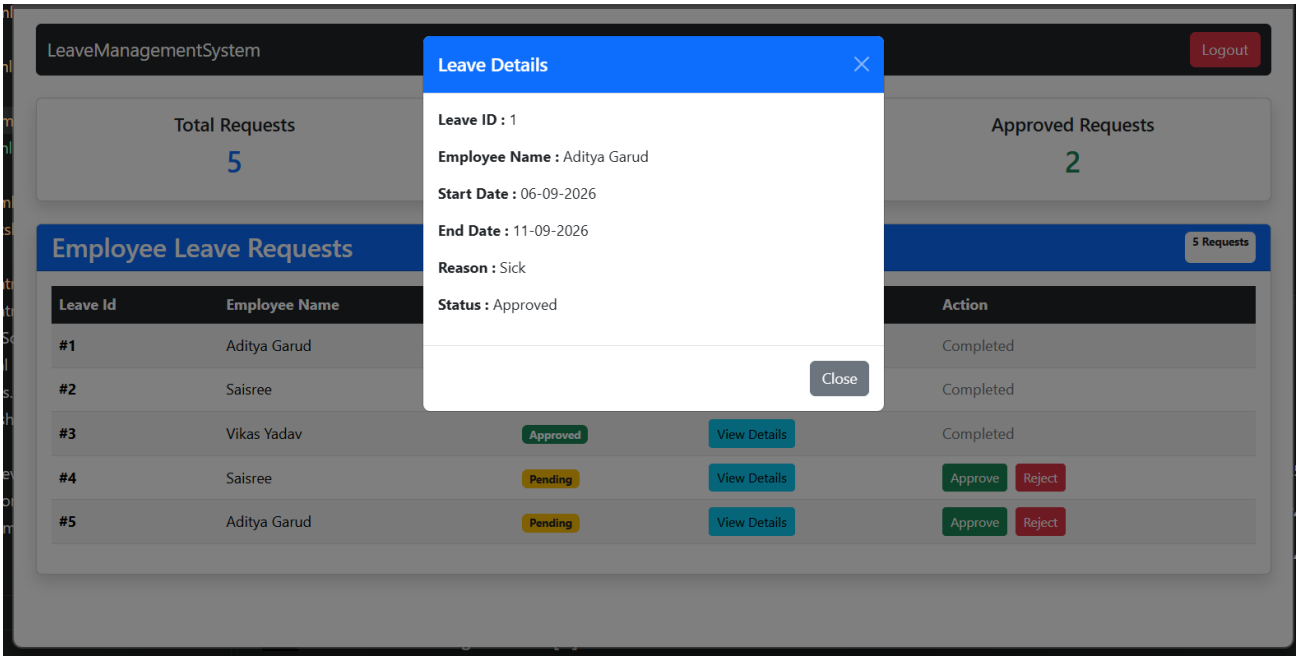


Figure 12: Leave Summary Page

The screenshots below illustrate the features available in the Admin Module of the Leave Management System. Administrators can view all employee leave requests, review leave application details, approve or reject leave requests, and monitor overall leave activities through a dedicated dashboard. The module also provides access to leave request summaries and helps maintain employee leave records in an organized manner. These features improve the efficiency of the leave approval process and provide a centralized platform for managing leave-related activities within the organization.

Development Phase

The development phase involved implementing all planned functionalities using ASP.NET Core MVC, Entity Framework Core, MySQL, and Bootstrap. The project was developed following the MVC architecture to ensure maintainability, scalability, and proper organization of application components.

Separate functionalities were implemented for Employees and Administrators based on their assigned roles.

Functionalities Implemented

Authentication and Authorization

- Session-Based Authentication
- Role-Based Authorization
- Login Validation
- Secure Access Control

Employee Management

- Leave Application Submission
- View Leave Requests
- Leave Status Tracking
- Leave Balance Management

Leave Management

- View Leave Requests
- Approve Leave Requests
- Reject Leave Requests
- Update Leave Status

The development phase successfully transformed the project requirements into a fully functional Leave Management System capable of handling employee leave activities efficiently.

Learning Outcomes and Conclusion

During the development of the Leave Management System, practical knowledge was gained in various technologies and software development concepts.

Technologies Learned

- ASP.NET Core MVC
- C#
- Entity Framework Core
- MySQL
- Bootstrap
- HTML
- CSS

Concepts Learned

- MVC Architecture
- Authentication
- Authorization
- Session Management
- CRUD Operations
- Database Relationships
- Entity Framework Core
- Software Development Life Cycle (SDLC)

The project provided valuable hands-on experience in database management, web development, and implementing business requirements through software solutions.

Conclusion

Staff Leave Manager - Leave Management System is a mini project developed to simplify and automate the leave management process within an organization. The application provides a centralized platform where employees can apply for leave, view leave requests, track leave status, and check their leave balance.

The system also enables administrators to review, approve, or reject leave applications while monitoring employee leave activities through an administrative dashboard. This helps reduce manual work and improves the overall efficiency of leave management.

The project was developed using ASP.NET Core MVC, Entity Framework Core, MySQL, and Bootstrap while following the Waterfall SDLC Model. The application successfully achieved its objective of providing a simple, secure, and user-friendly solution for managing employee leave requests.

Through the development of this mini project, valuable practical experience was gained in web application development, database management, authentication, authorization, and MVC architecture. Overall, the Leave Management System serves as an effective academic project that demonstrates the implementation of modern software development technologies and leave management automation.

Thank You.